

This diagram illustrates the electrical connections for the RykerConnect module, showing various components and their interconnections.

Controller and connectors

J6 ESP32 socket P1

PIN	SIGNAL	WAVESHARE
1	NC_1	
2	NC_2	
3	NC_3	
4	NC_4	
5	NC_5	
6	NC_6	
7	GPI07	
8	DSP_RES	
9	NC_9	
10	NC_10	
11	SCL	
12	SDA	
13	NC_13	
14	NC_14	
15	NC_15	
16	CS2	
17	SPL_MOSI	
18	SPL_SCK	
19	DSP_DC	
20	CS1	
21	+5V_MODULE	
22	GND	

J7 ESP32 socket P2

PIN	SIGNAL	WAVESHARE
1	GND	
2	NC_2	
3	NC_3	
4	NC_4	
5	NC_5	
6	NC_6	
7	NC_7	
8	NC_8	
9	NC_9	
10	NC_10	
11	NC_11	
12	NC_12	
13	NC_13	
14	NC_14	
15	NC_15	
16	NC_16	
17	NC_17	
18	NC_18	
19	NC_19	
20	NC_20	
21	GND	
22	GND	

J3 SMOBB-SR5S-TB(LF)(SN)

PIN	SIGNAL	WAVESHARE
1	Pin_1	GND
2	Pin_2	+3.3V
3	Pin_3	SPL_SCK
4	Pin_4	SPL_MOSI
5	Pin_5	DSP_RES
6	Pin_6	DSP_DC
7	Pin_7	CS1
8	Pin_8	CS2
MP	MP	GND

J5 SMO4B-SR5S-TB(LF)(SN)

PIN	SIGNAL	WAVESHARE
1	Pin_1	GND
2	Pin_2	+3.3V
3	Pin_3	SDA
4	Pin_4	SCL
MP	MP	GND

R3 4.7k 1%

PIN	SIGNAL	WAVESHARE
1	Pin_1	+3.3V
2	Pin_2	SCL

R4 4.7k 1%

PIN	SIGNAL	WAVESHARE
1	Pin_1	+3.3V
2	Pin_2	SDA

J1 436500200

PIN	SIGNAL	WAVESHARE
1	Pin_1	+12V
2	Pin_2	GND

Waveshare 28836 socketed module. ESP 3.3V header pins intentionally isolated. Peripherals use U7. Disconnect USB before closing J8 for external power.

Accessory protection

File: sheet2.kicad_sch

5V regulator and USB isolation

File: sheet3.kicad_sch

Battery-backed clock and light sensor

File: sheet4.kicad_sch

Sheet: /		Rev: REV05-12V	
Size: A3	Date: 2026-09-13	Id: 1/4	
KiCad E.D.A. 10.0.6			

Waveshare 28836 socketed module, ESP 3.3V header pins intentionally isolated.
Peripherals use U7. Disconnect USB before closing JB for external power.

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File: sheet2.kicad_sch

File: sheet3.kicad_sch

File: sheet4.kicad_sch

Sheet: /
File: RykerConnect.kicad_sch

Title: RykerConnect – Controller and connectors

Size: A3	Date: 2026-09-13	Rev: REV05-12V
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Rev: REV05-12V

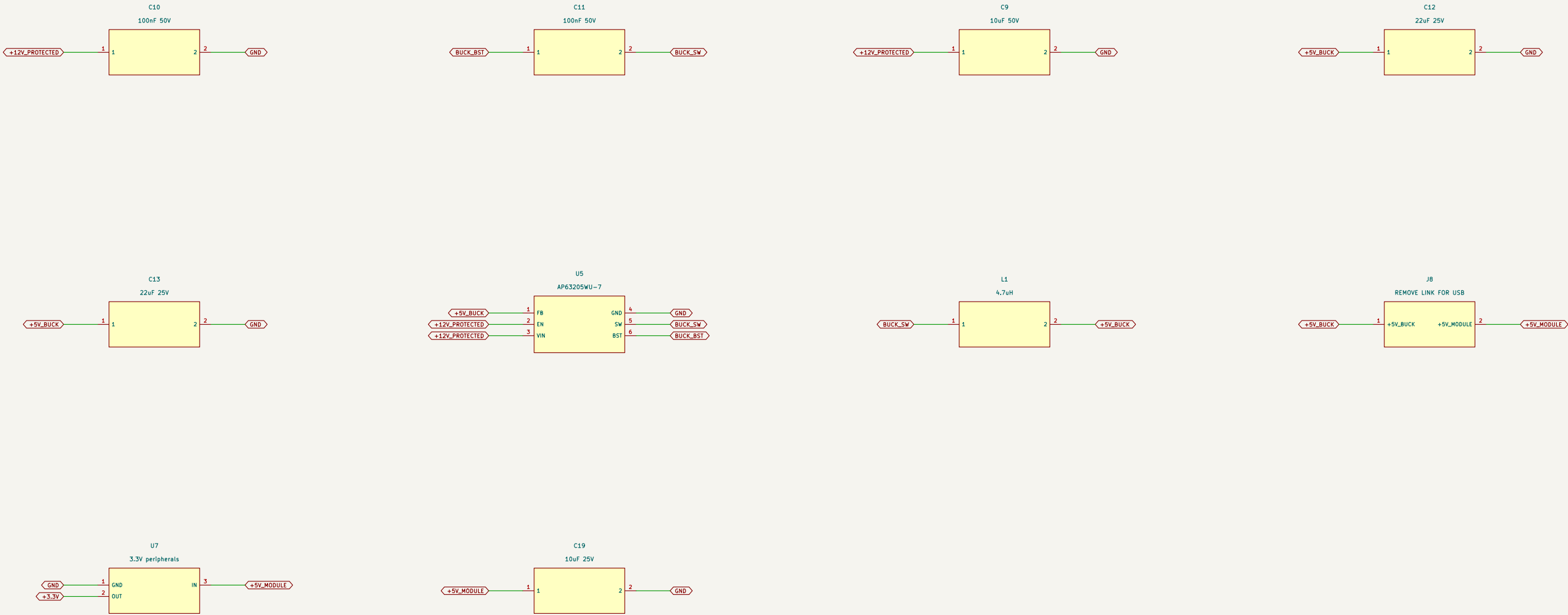
Id: 1/4

Accessory protection



12V variant only: fused accessory input. RTN is isolated from GND.
USB variant: vehicle protection is external to this board.

5V regulator and USB isolation



Battery-backed clock and light sensor

PCF8563 at I2C 0x51. D5/D6 isolate the non-rechargeable CR2032.
X1 plus C18 form the 32.768kHz oscillator; verify drift on prototype.
Remote BME280 provides temperature/humidity/pressure; no local MCP9808.

Sheet: /Battery-backed clock and light sensor/ File: sheet4.kicad_sch		Rev: REV05-12V	
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Battery-backed clock and light sensor

BT1
BAT-HLD-001

J4
SM02B-SRSS-TB(LF)(SN)

C3
100nF 50V

C6
100nF 50V

C5
10uF 25V

R6
10k 1%

U3
PCF8563T/5.518

X1
32.768kHz 12.5pF

C18
22pF COG

D5
BAV199 backup isolation

D6
BAV199 backup isolation

PCF8563 at I2C 0x51. D5/D6 isolate the non-rechargeable CR2032.
X1 plus C18 form the 32.768kHz oscillator; verify drift on prototype.
Remote BME280 provides temperature/humidity/pressure; no local MCP9808.

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